

SHAHRZAD AMIN RANJBAR

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DATA SCIENTIST | AI/ML ENGINEER

Astute and analytical professional with 4 years of cutting-edge industry experience, specializing in AI solutions for banking. Proven ability to turn data into revenue-driving models using Python, Spark, and Azure. Expert deploying ML at scale, driving first-rate strategy, and ensuring models are explainable, ethical, and production-ready.

BUSINESS & TECHNICAL SKILLS:

Prediction Service | Advanced EDA | Predictive Data Analytics | Cross-Functional Communication | Corporate Compliance
ML & Deep Learning: Supervised & Unsupervised Learning | Ensemble Methods | Neural Networks | NLP | Time Series Forecasting

Data Tools: Pandas | NumPy | Matplotlib | MLflow | Git | Spark | LangChain

Platforms: Databricks | Microsoft Azure | Snowflake

Languages: Python | SQL

MLOps: CI/CD, versioning, testing, and monitoring of ML models in production

PROFESSIONAL EXPERIENCE

Ernst & Young LLP (EY) November 2025 - Present Senior Consultant - Data Scientist

- Led and developed an AI-driven payroll intelligence and validation framework for UPS California ET EY, leveraging Retrieval-Augmented Generation (RAG) to extract and operationalize complex union pay rules from unstructured agreements, integrating them with large-scale payroll transaction data through scalable data pipelines and explainable AI to automatically detect compensation discrepancies, calculation errors, and compliance risks—significantly improving payroll accuracy and reducing financial leakage.

Business Development Bank of Canada (BDC) October 2022 – June 2025 Data Scientist

- Developed a centralized AI Sales Companion using RAG and LangChain to integrate real-time client data with transformer models LLM, enabling dynamic profiling, historical insights, and predictive analytics—accelerating sales decisions and driving \$1M in new revenue within 3 months.
- Developed and deployed a prediction service enabling Business Development Leads to accurately identify client segments with high repeat business potential, contributing ~\$3M in annual revenue. Conducted advanced EDA and feature engineering on multi-domain banking data, designed data pipelines and target definitions, and used LightGBM with hyperparameter tuning and MLflow for model management. Ensured model interpretability for stakeholder clarity.
- Integrated fairness, transparency, and bias mitigation techniques into a real-world AI system, improving model accountability and reducing bias-related errors by ~30%, ensuring ethical AI practices and responsible data use.
- Engineered a robust data aggregation pipeline and applied machine learning for deep data analysis—streamlining data structuring and improving insight extraction speed by ~3x, driving ≥15% performance gains.

Inmindtechnologies Inc. October 2021 – June 2022 Data Scientist

- Developed and deployed an order completion prediction service for Bell Canada, processing over 10M structured records. Applied advanced data cleansing, engineered 20+ predictive features, and tuned XGBoost and LightGBM models, improving prediction accuracy by 22% and reducing order fallout costs by an estimated \$2M annually.

- Predicted an anomaly detection model on 3D printing AIoT for Additive manufacturing. Leveraged data wrangling, data augmentation, and data scaling and modeling of MobilenetV2 and Vanilla convolutional neural networks (CNNs), resulting in publishing an article in ICAM 2021.
- Directed data analysis, including data wrangling, data cleaning methods, data preparation, manipulation, and feature engineering on a vast but well-organized dataset, utilized various visualization techniques to analyze the weekly and monthly income rate compared to the number of projects during the COVID period.

Concordia University July 2019 – September 2020 Graduate Research Assistant

- Implemented face mask detection from scratch in Python, leveraging data preparation, PyTorch Lightning, and convolutional neural networks. Analyzed model performance with and without cross-validation, observing a 25% increase in cross-entropy when detecting masked faces.
- Constructed a speech recognition project for trigger/wake word detection. Synthesized and processed audio recordings to create train/dev datasets, trained a trigger word detection by RNN models, and made a prediction that can be run, with 90% accuracy, on the laptop so that every time “activate” is said, it starts up your favorite task.

EDUCATION

Master of Engineering (MEng), Master of Computer Science (MCS)

Concordia University

Bachelor of Engineering (BEng), Bachelor of Software Engineering

Azad University

CERTIFICATION

- Generative AI with Large Language Models, Deeplearning.ai & AWS, July 2025.
- Retrieval Augmented Generation (RAG), Deeplearning.ai, Sep 2025
- Generative AI Fundamentals, Databricks, April 2025.
- Trustworthy and Responsible AI, Mila, January 2024.
- Neural Networks & Deep Learning (deep learning specialization), Deeplearning.ai, February 2021.
- Improving deep Neural Network (Hyperparameter tuning, regularization, and optimization), Deeplearning.ai, June 2021.
- Machine Learning Engineering for Production (MLOps) Specialization, Deeplearning.ai, September 2020.
- Sequence Models (RNN, NLP, deep learning specialization), Deeplearning.ai, October 2020.

